SCOPE OF WORK
Centralized Coal Logistics & Penalty Control System

1. PROJECT OVERVIEW

The objective of this project is to design, develop, and implement a **Centralized Coal Logistics & Penalty Control System** to replace Excel-based data handling at three railway sidings:

- **Pakur**
- **Dumka**
- **Kurwa**

The system shall digitally manage the **end-to-end coal movement lifecycle** from:
Mine → Road Dispatch → Railway Siding → Rake Loading → In-motion Weighment → Railway Receipt (RR) → Power Plant → Reconciliation & MIS
The application will act as a **single source of truth**, where operational teams only enter data once, and all downstream calculations, validations, alerts, penalties, and dashboards are handled automatically by the system.

2. KEY OBJECTIVES

- Minimize railway penalties (especially overloading & demurrage)
- Enable **real-time penalty prediction before RR**
- Provide **consolidated dashboards** for management
- Eliminate Excel-based duplication and manual analysis
- Improve operational discipline through alerts and accountability

3. USER ROLES & ACCESS CONTROL

- Mine Operator
- Siding Operator
- Siding In-Charge
- Railway Cell
- Management
- System Admin

Role-based access and edit permissions mandatory.

4. FUNCTIONAL SCOPE (MODULE-WISE)

4.1 MINE → SIDING ROAD DISPATCH MODULE

Purpose: Digitize coal dispatch from mines to railway sidings.
Developer Scope:

- Vehicle authorization entry
- Gate pass / loading slip tracking (previous trip dependency)
- Weighbridge data entry (gross / tare / net)
- JIMMS challan reference capture
- GPS trip reference (manual reference only; no GPS integration in Phase-1)
- Trip status tracking (Dispatched / In Transit / Received)

Key Outputs:

- Truck-wise DPR
- Mine dispatch summary

Note: the above-mentioned module is already existed in our existing SHAReTrack application.

4.2 SIDING COAL RECEIPT & STOCK MODULE

Purpose: Maintain live siding stock based on truck receipts.

Developer Scope:

- Shift-wise coal receipt entry
- Vehicle-wise receipt confirmation
- Auto update of siding stock
- Stock availability check for rake planning

Validations:

- Cannot receive vehicle without dispatch reference
- Cannot raise indent without sufficient stock

4.3 RAKE INDENT & PLANNING MODULE

Purpose: Digitize indent raising based on stock availability.

Developer Scope:

- Indent creation screen
- Capture indent date, time, siding, target quantity
- Link indent to siding stock
- Railway reference number field

System Logic:

- Indent is **stock-driven**, not date-driven
- Indent status tracking (Raised / Acknowledged / Closed)

4.4 RAKE PLACEMENT & TXR MODULE

Purpose: Capture critical pre-loading events.

Developer Scope:

- Rake placement time capture
- TXR inspection start & end time entry
- Auto TXR duration calculation
- Unfit wagon count

Outputs:

- TXR duration report
- Fit vs unfit wagon status

4.5 UNFIT WAGON MANAGEMENT MODULE

Purpose: Track and exclude unfit wagons from loading.

Developer Scope:

- Wagon-wise unfit entry
- Reason for unfit
- Marking method (flag / reflector / light)
- Auto exclusion from loading screens

4.6 WAGON-WISE LOADING & LOADER DATA MODULE (CRITICAL)

Purpose: Capture and analyze loader-wise loading data.

Developer Scope:

- Wagon-wise loading entry screen
- Loader ID and operator mapping
- CC capacity reference

- Loaded quantity entry
- Support for:
    - Loader application data (future integration)
    - Manual log book entry

System Logic:

- Auto flag overload / underload risk
- Loader-wise performance tracking

4.7 IN-MOTION WEIGHMENT MODULE

Purpose: Capture actual railway weighment data.

Developer Scope:

- Wagon-wise in-motion weighment entry
- Slip number upload / reference
- Time capture

4.8 LOADER vs WEIGHMENT COMPARISON ENGINE

Purpose: Predict penalties before RR.

Developer Scope:

- Automatic comparison of:
    - Loader-noted quantity
    - In-motion weighment quantity
- Difference calculation
- Overload / underload classification
- Alert generation

Key Output:

- Early warning dashboard (Pre-RR)

4.9 LOADING TIME & RAKE MOVEMENT MONITORING

Purpose: Prevent demurrage.

Developer Scope:

- Auto start 3-hour free time counter from rake placement
- Live time balance display
- Loading completion time capture
- Rake movement permission & actual movement time
- Delay calculation
- Alert if delay exceeds thresholds

4.10 RAILWAY RECEIPT (RR) MODULE

Purpose: Capture final commercial authority.

Developer Scope:

- RR upload (PDF)
- Manual RR data entry fields
- Freight, penalty, GST capture
- Link RR to rake

Important Rule:

- RR is final financial authority
- RR data cannot be edited after submission

4.11 RECONCILIATION ENGINE
Purpose: End-to-end quantity and penalty reconciliation.
Developer Scope:

- Mine vs Siding
- Siding vs Rake
- Rake vs Weighment
- Weighment vs RR
- RR vs Power Plant receipt

4.12 PENALTY MANAGEMENT MODULE
Purpose: Consolidated penalty visibility.
Developer Scope:

- Penalty type classification
- Pre-RR predicted penalty
- Post-RR actual penalty
- Siding-wise, rake-wise penalty register

4.13 DASHBOARDS & MIS
Purpose: Management decision support.
Developer Scope:

- Siding-wise dashboard (Pakur / Dumka / Kurwa)
- Date-wise dispatch & penalties
- Rake-wise performance
- Loader-wise overloading trends
- Power plant–wise dispatch

FINAL NOTE FOR DEVELOPER:
This system is **not a data-entry application**, but an **operational control and penalty prevention system**. Accuracy of timestamps, wagon-level data integrity, and alert logic are critical. The developer is expected to strictly follow the defined workflows and validations without altering business logic.

End of Scope of Work